QUANTUM ERROR CORRECTION (QEC) — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: April 15, 2026

Diagram-based questions on Quantum Error Correction (QEC)

Question 1

10.0 points

Examine the circuit shown below of a stabilizer code. Identify how the syndrome measurement aids in error detection and explain the role of stabilizers in quantum error correction.

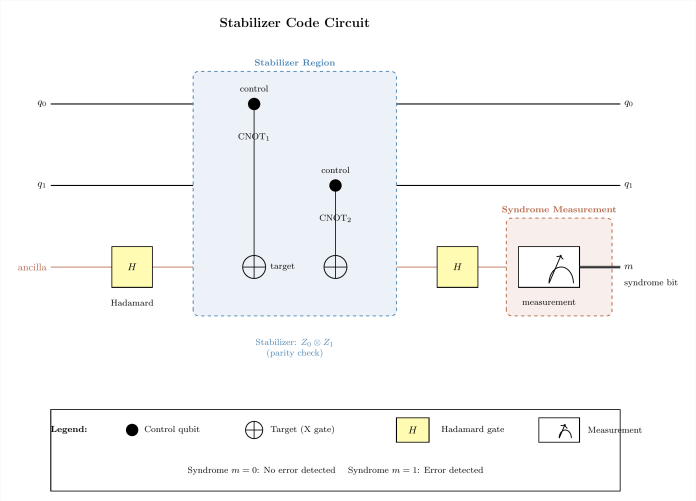

Fig. 1. Quantum circuit diagram for a stabilizer code. Include horizontal lines for qubit wires labeled q0, q1, and ancilla. Add CNOT gates connecting q0 to ancilla and q1 to ancilla. Include H gate on ancilla qubit. Add measurement gates at the end of ancilla qubit line. Highlight the stabilizers and syndrome measurement components in the circuit.

Question 2

10.0 points

Examine the circuit shown below illustrating the 3-qubit bit-flip code with its corresponding error detection circuit. Explain the error correction process and identify the role of each circuit component in detecting a single bit-flip error.

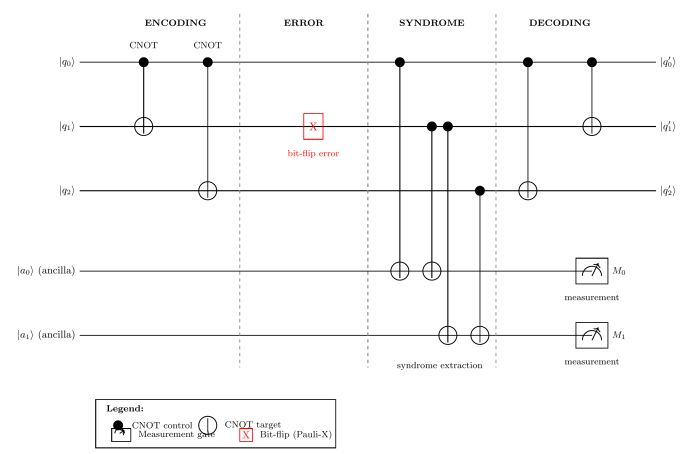

Fig. 2. A quantum circuit illustrating the 3-qubit bit-flip code with its corresponding error detection circuit. The circuit includes three qubit lines labeled q0, q1, and q2. Each qubit line has a series of CNOT gates for encoding and decoding. The circuit also includes ancilla qubits for error detection, with measurement gates at the end of the ancilla lines. The CNOT gates are used to entangle the qubits, and the measurement gates are used to detect errors. The circuit should clearly label each qubit line and gate type (CNOT, measurement).

Question 3

10.0 points

Examine the surface code lattice in the diagram below. Describe how logical qubits are encoded on the lattice and discuss how the layout is used for error correction.

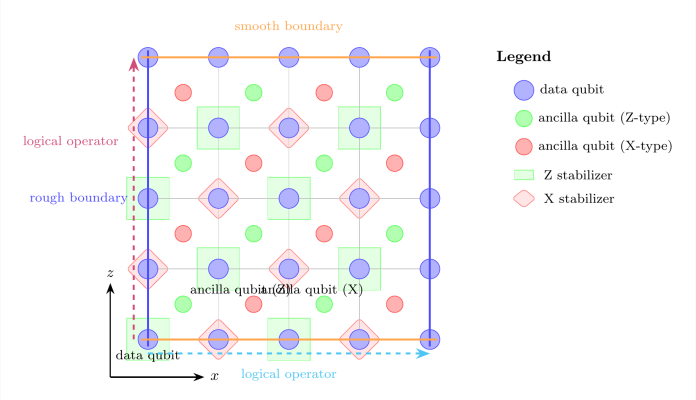

Fig. 3. A surface code lattice diagram showing a grid of qubits. Each qubit is represented as a circle, with data qubits and ancilla qubits clearly labeled. The lattice should include both X and Z stabilizers, represented as squares and stars, respectively, connecting groups of qubits. The diagram should illustrate the arrangement of qubits in a 2D grid, highlighting the pattern used for encoding logical qubits and performing error correction. Include labels for logical operators and any relevant boundaries or defects in the lattice. IMPORTANT: The diagram is missing required labels. The diagram is missing the required labels for "logical operators. Add each missing label as visible text at the appropriate location in the diagram. IMPORTANT correction needed: The diagram explicitly labels the "logical operators", which are. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only.

Question 4

10.0 points

Examine the diagram below illustrating fault-tolerant gate teleportation. Describe how teleportation is utilized to achieve fault-tolerance in quantum gates and what protections it offers against errors.

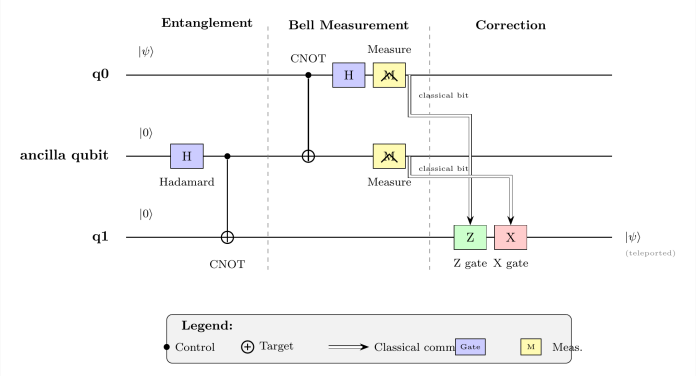

Fig. 4. A quantum circuit diagram illustrating fault-tolerant gate teleportation. The diagram should include three qubit lines labeled q0, q1, and ancilla. The circuit should show an initial entanglement operation between q0 and ancilla using a CNOT gate, followed by a measurement gate on q0. The measurement result should be classically communicated to q1, where a corrective operation (X or Z gate) is applied based on the measurement outcome. The circuit should also include a final CNOT gate between ancilla and q1 to complete the teleportation process. Label all gates and qubit lines clearly. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram shows a standard quantum teleportation. Regenerate using EXACTLY the values and sequences specified. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram shows a standard quantum teleportation circuit. Regenerate using EXACTLY the values and sequences specified.

Question 5 10.0 points

Interpret the threshold theorem in the diagram below, which illustrates error rates in quantum error correction. Explain how the error rate threshold is defined and its significance in achieving fault-tolerant quantum computation.

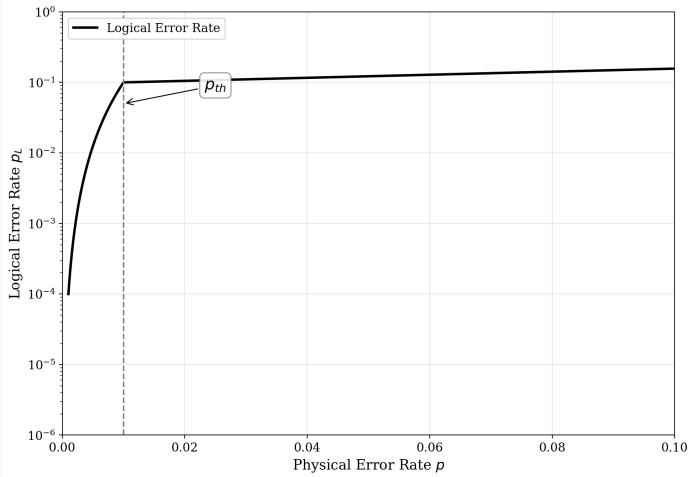

Fig. 5. CRITICAL: The previous diagram showed the wrong type. The diagram shows a single curve where the logical error. Regenerate from scratch showing the CORRECT diagram type. CRITICAL: The previous diagram showed the wrong type. The diagram shows a plot of "Error Rate. Regenerate from scratch showing the CORRECT diagram type. A plot illustrating the error rate threshold in quantum error correction. The x-axis represents the physical error rate, while the y-axis shows the logical error rate. The plot should include a curve that demonstrates the threshold behavior, where below a certain physical error rate, the logical error rate decreases significantly, indicating the threshold for fault-tolerant quantum computation. Include labels for the axes and the threshold point.